Network Science

Lecture 07: Small-World Networks

5942UE Network Science

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Small-World Networks

This lecture asks why short paths are common and when decentralized navigation is possible.

Focus

- the small-world phenomenon
- Watts–Strogatz structure and randomness
- decentralized search
- empirical validation and the Web as a directed information network

Module 1

Small-World Phenomenon

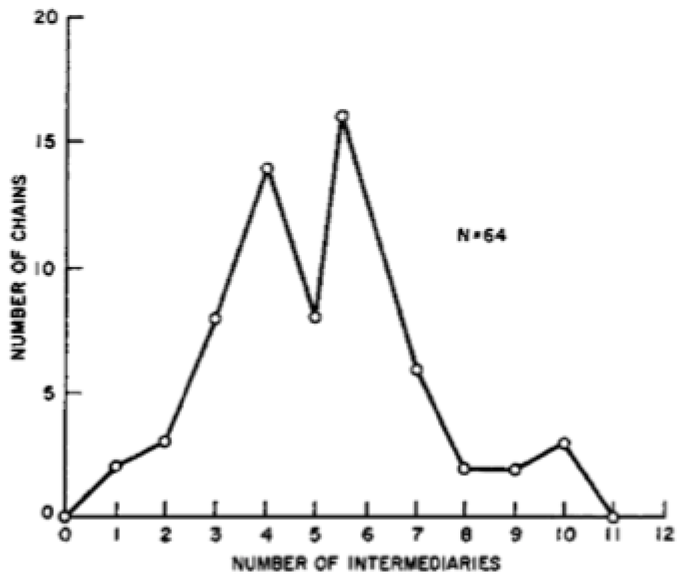
Guiding Question

Why are distances in large networks often surprisingly short?

Formal Intuition

The small-world claim is that typical shortest-path distances grow **slowly** relative to network size.

Milgram



N=64
NUMBER OF CHAINS
NUMBER OF INTERMEDIARIES
0
5
10
15
20
0 1 2 3 4 5 6 7 8 9 10 11 12

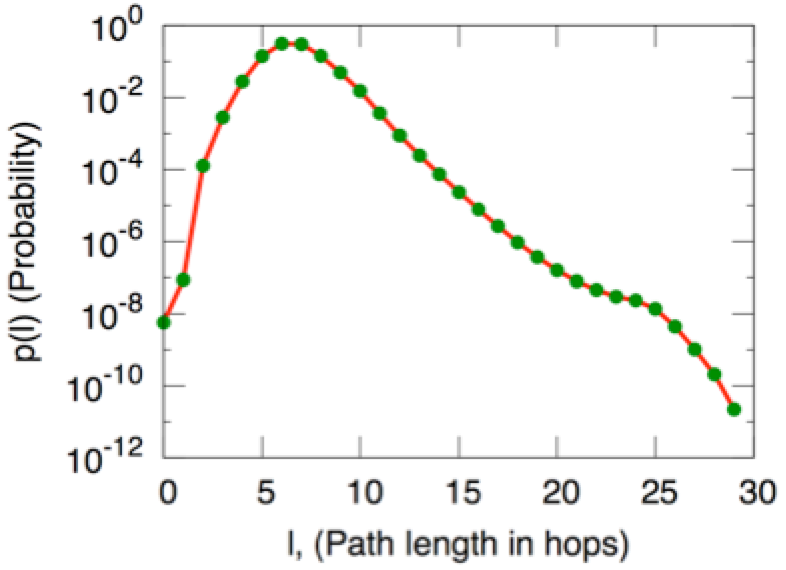
Core Result

$$d_{\text{typical}} \propto \log |V|$$

Interpretation

The small-world claim is **not** that everyone is directly connected. It is that typical distances grow slowly even in very large networks.

Examples



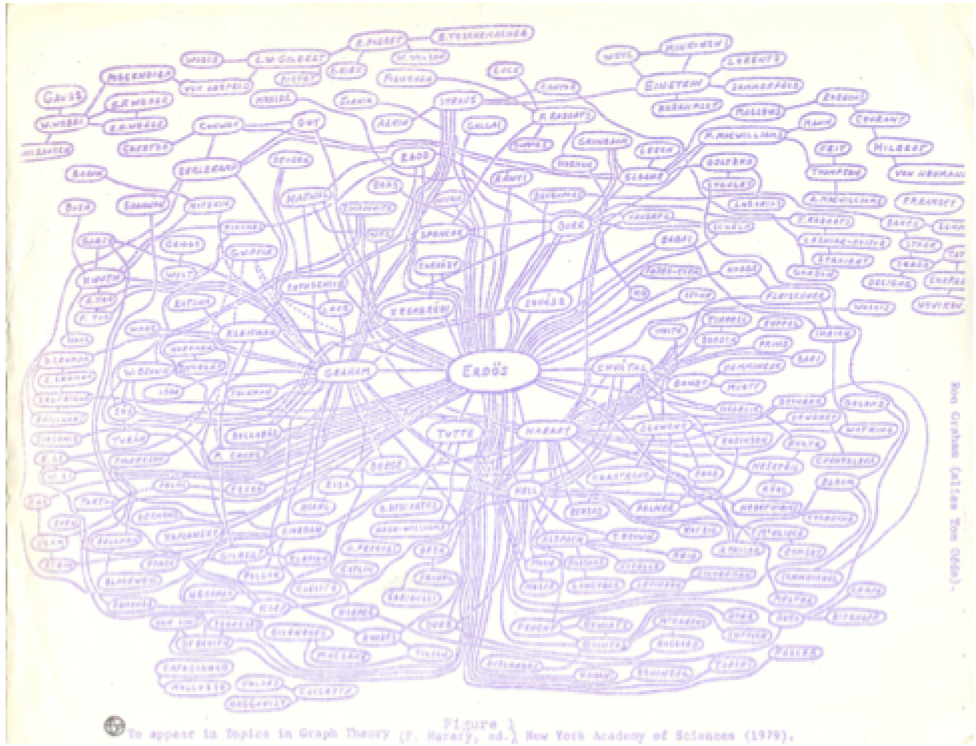


Figure 2.12: Ron Graham's hand-drawn picture of a part of the mathematics collaboration graph, centered on Paul Erdös [189]. (Image from http://www.oakland.edu/enp/cgraph.jpg)

> Large networks can still be navigably small in terms of path length.

Module 2

Structure and Randomness

Guiding Question

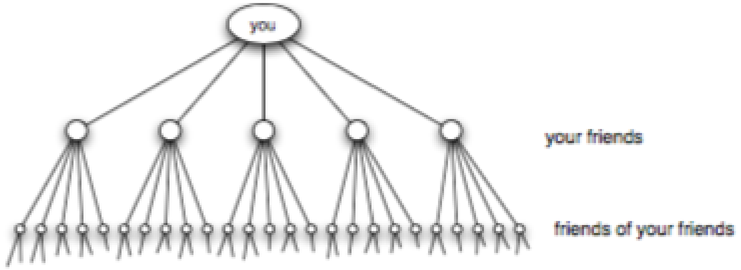
How can a network be both highly clustered and still have short paths?

Formal Intuition

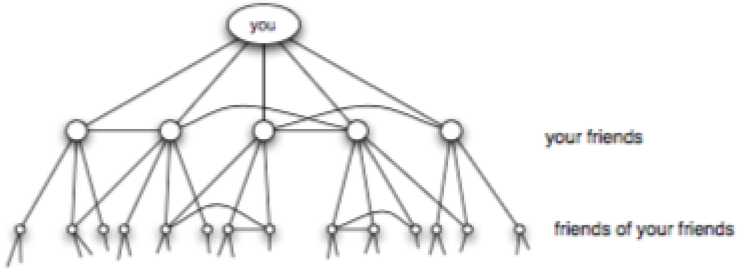
Clustering and short average path length are **not mutually exclusive**.

The key is that a few **long-range links** can drastically reduce distances.

Formation of Short Paths

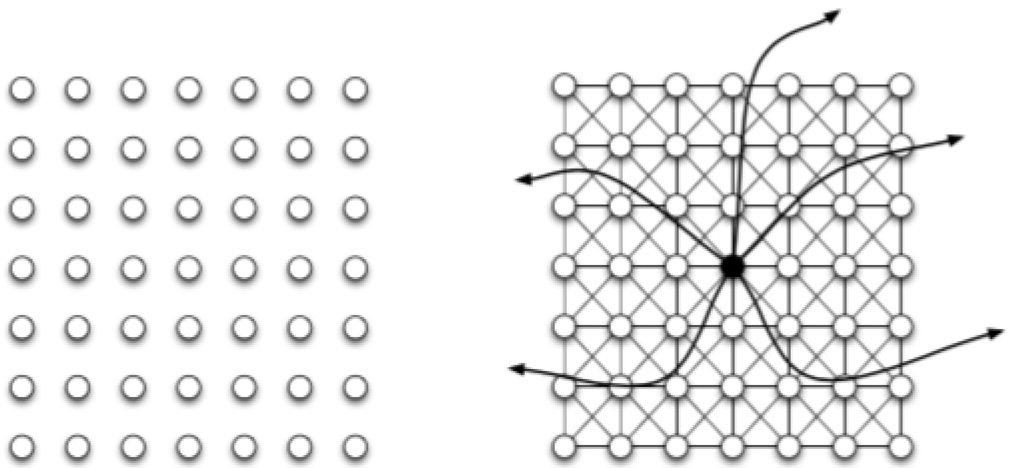


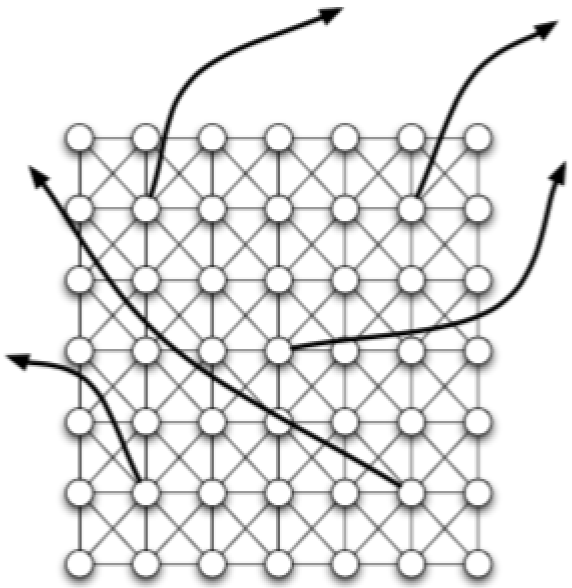
(a) *Pure exponential growth produces a small world*



(b) *Triadic closure reduces the growth rate*

Watts–Strogatz





Interpretation

Local links preserve **clustering.** A small amount of random long-range linkage collapses **path length** dramatically.

> A tiny dose of randomness can make the world small without destroying local structure.

Module 3

Decentralized Search

Guiding Question

Why do short paths not automatically imply that decentralized actors can find them?

Problem

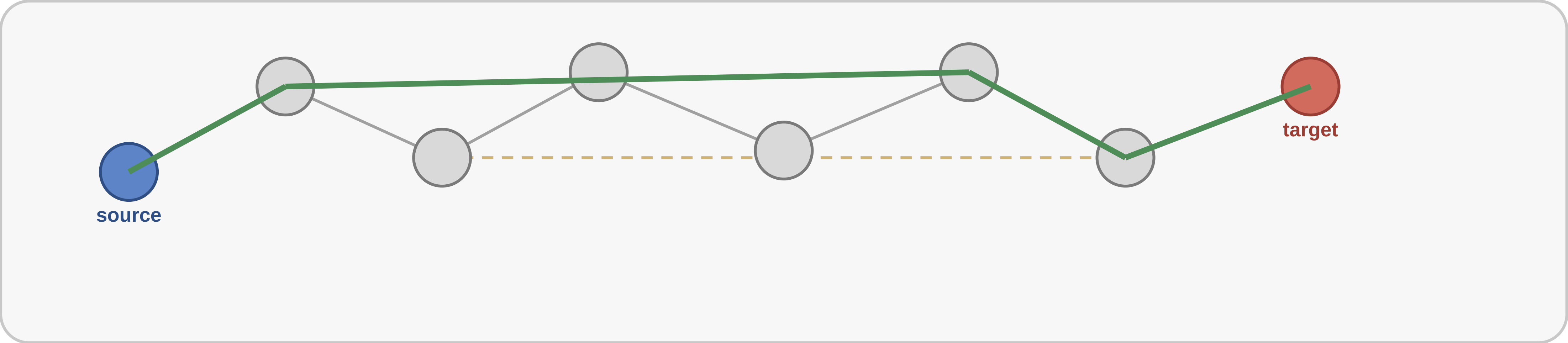
Milgram-like routing is **not** breadth-first search. Each actor sees only local information.

Formal Intuition

Decentralized search succeeds when the network places long-range links in a way that **local decisions can exploit**.

Kleinberg-Style Idea

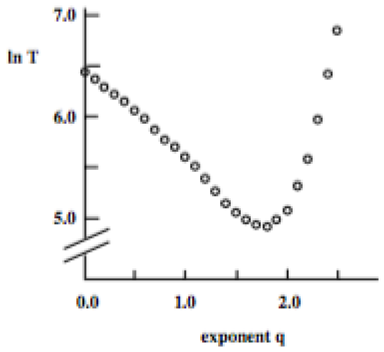
Decentralized search uses local views, not global shortest paths

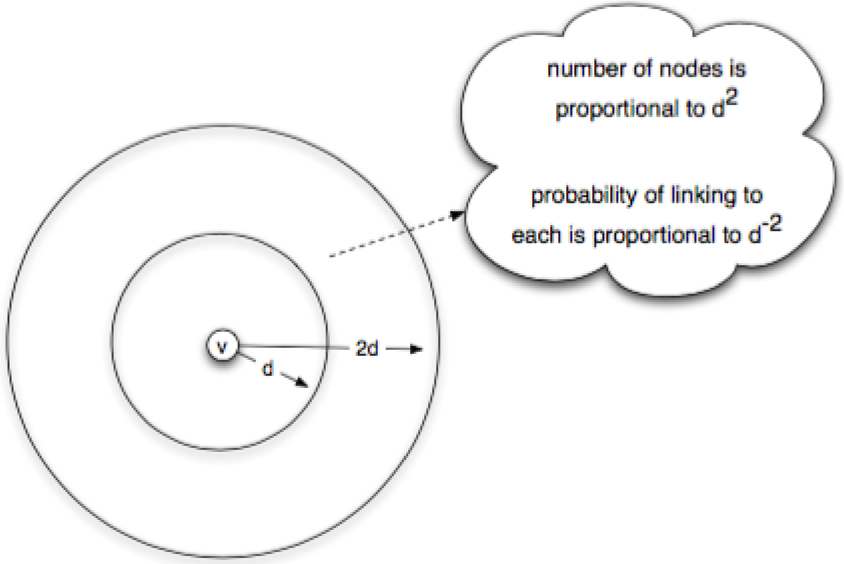


Green path is found by local forwarding; dashed links are long-range shortcuts that make local search possible.

Long-range links should not be uniformly random. They should be distributed **across scales**.

Evidence





> Efficient search requires not only short paths, but the right geometry of long-range ties.

Module 4

Empirical Analysis

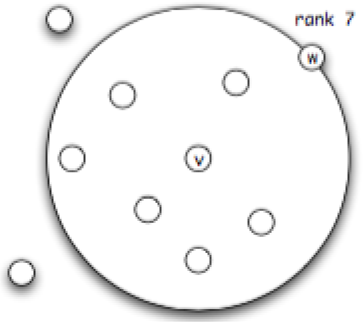
Guiding Question

Does the decentralized-search model show up in real data?

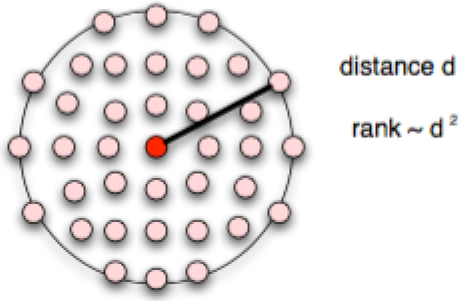
Formal Intuition

The empirical question is whether real networks organize long-range links across scales in the way search models predict.

Geographic Rank

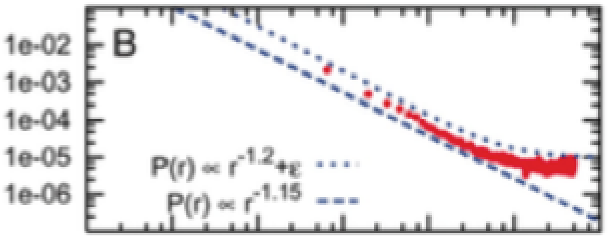


(a) *w is the* 7^{th} *closest node to v.*

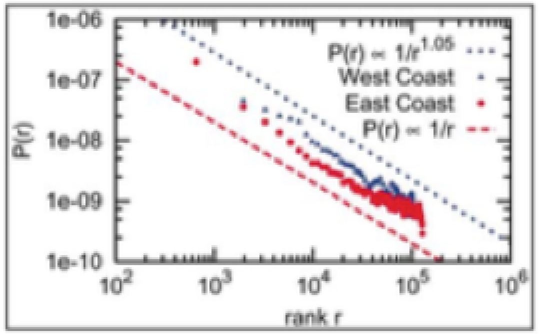


(b) *Rank-based friendship with uniform population density.*

Evidence

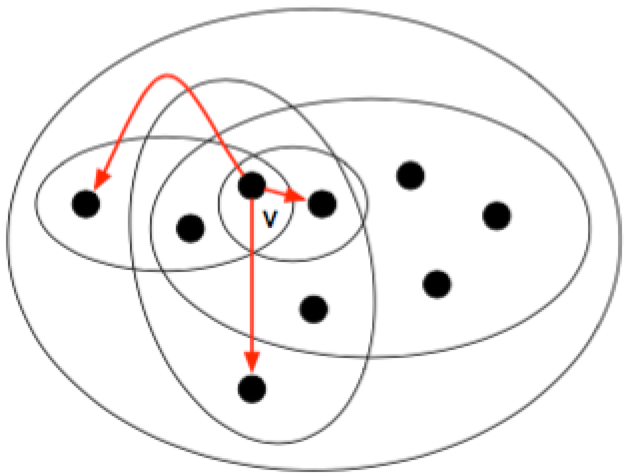


(a) *Rank-based friendship on LiveJournal*



(b) *Rank-based friendship: East and West coasts*

Beyond Geography



> Empirical networks often organize weak ties in ways that support decentralized navigation, even wher
> distance is social rather than geographic.

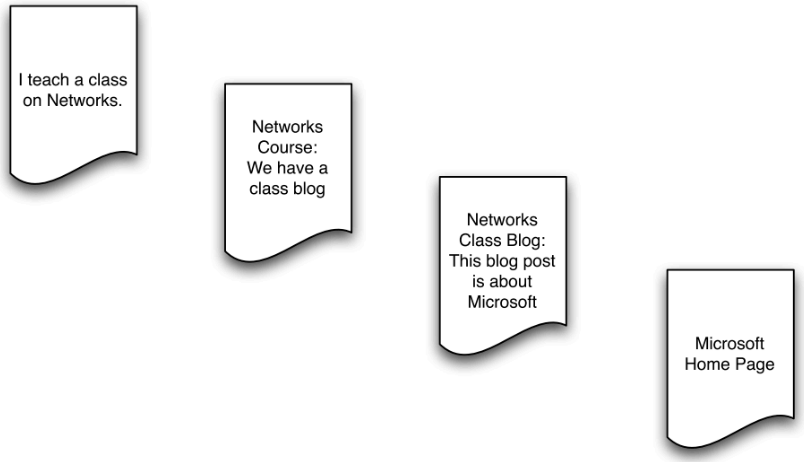
Module 5

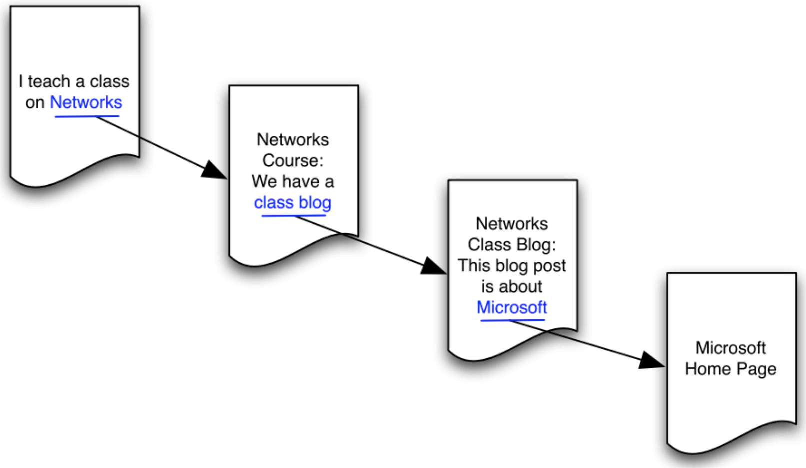
The Web as a Directed Network

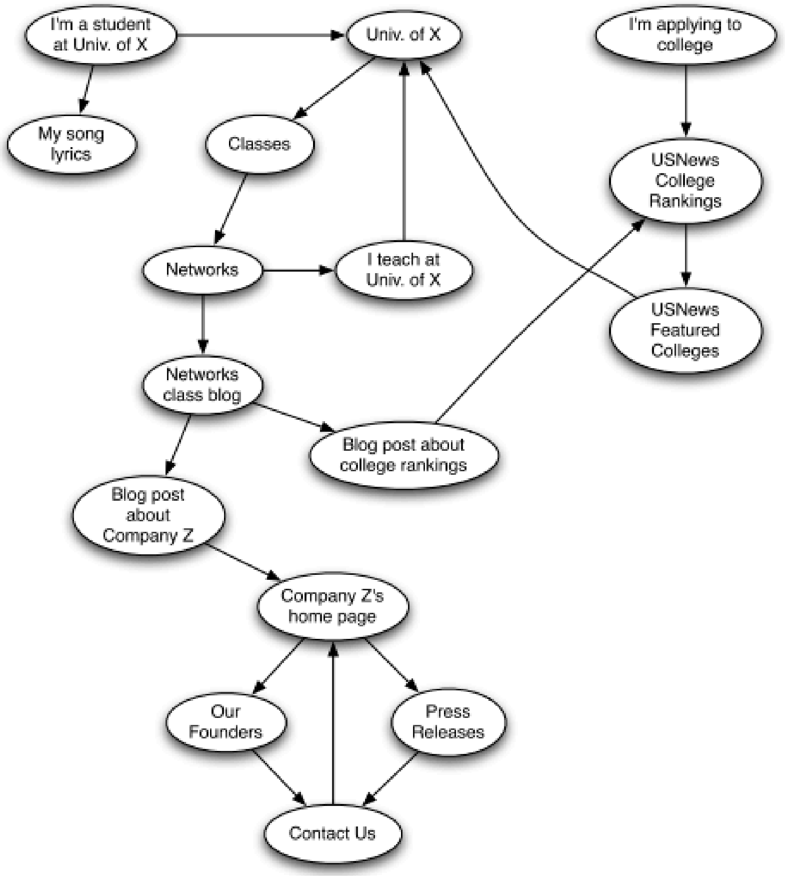
Guiding Question

What changes when we move from social networks to the Web?

From Hypertext to Directed Graphs

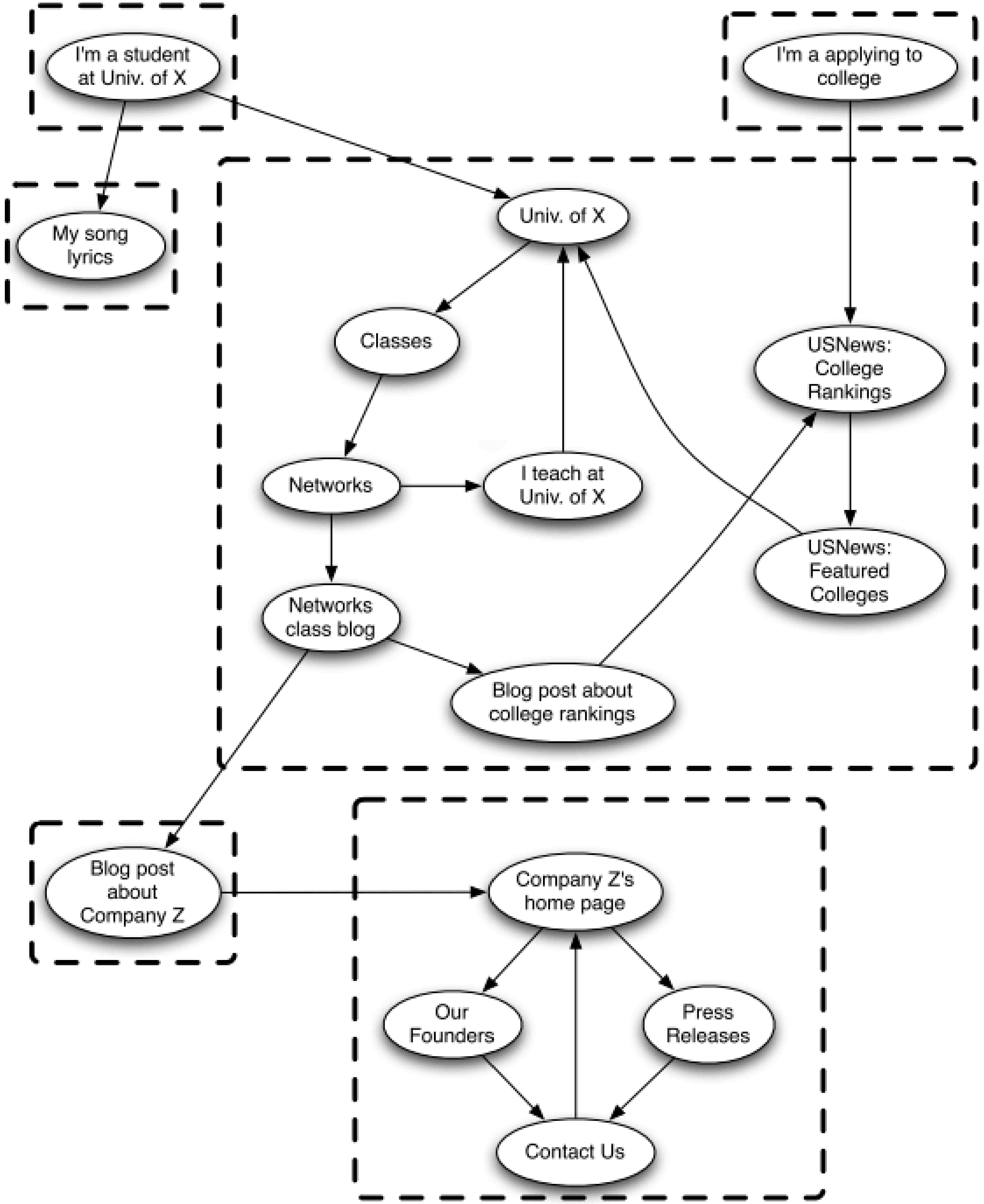




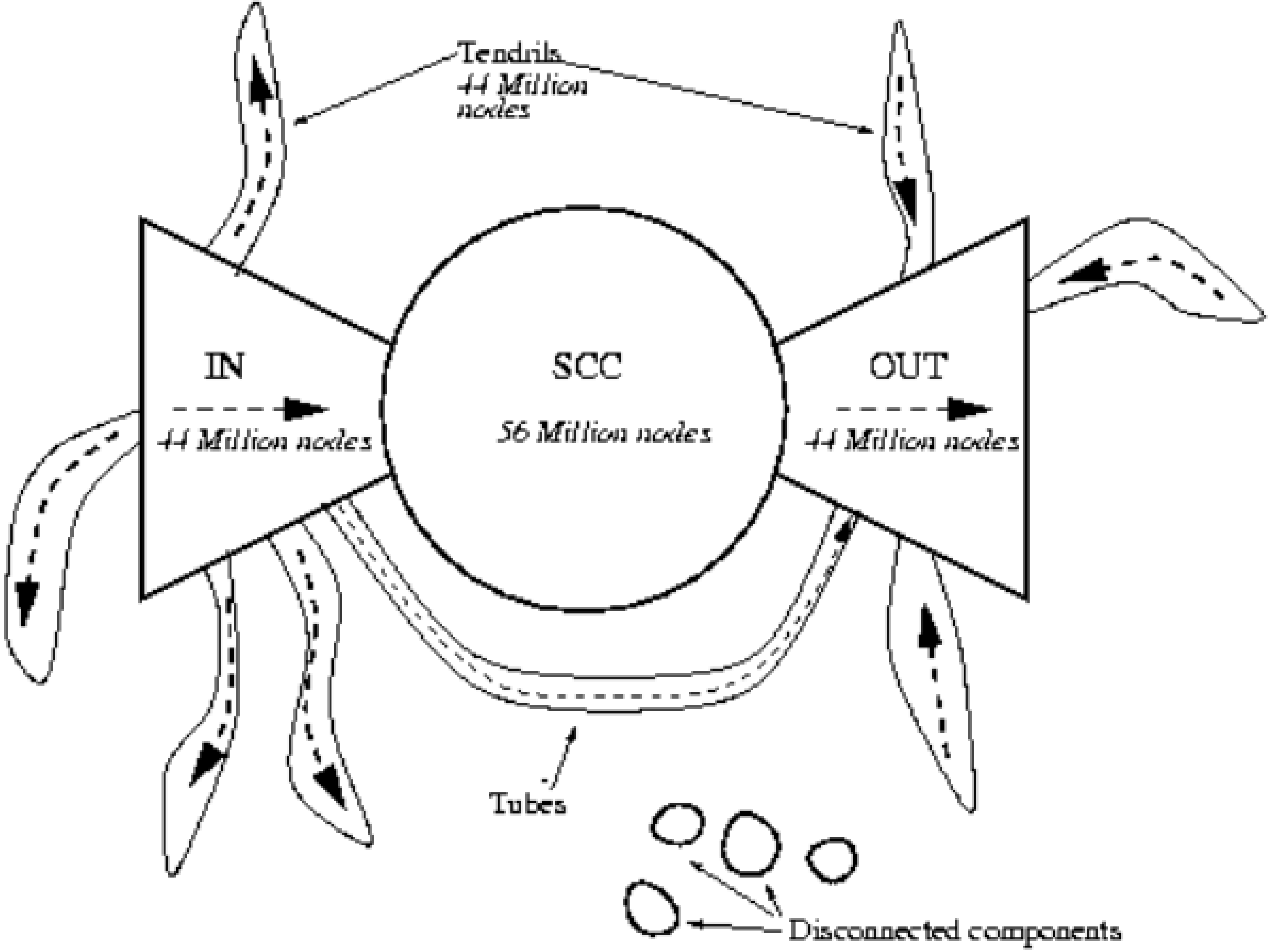


Strong Connectivity

In directed graphs, reachability is **asymmetric**, so connectedness must also be defined asymmetrically.


I'm a student at Univ. of X
I'm a applying to college
My song lyrics
Univ. of X
Classes
USNews: College Rankings
Networks
I teach at Univ. of X
USNews: Featured Colleges
Networks class blog
Blog post about college rankings
Blog post about Company Z
Company Z's home page
Our Founders
Press Releases
Contact Us

Bow-Tie Structure


Tendrils
44 Million
nodes
IN
44 Million nodes
SCC
56 Million nodes
OUT
44 Million nodes
Tubes
Disconnected components

Wrap-Up

- short paths alone do not guarantee navigability
- a little randomness can shrink distances dramatically
- information networks need directed notions of path, components, and reachability